

MAT 2235 Algebraic Structures
QUIZ – 4 Last Date for Submission: 19 April 2026
(Submit as single PDF attachment through LMS)

1. Consider $f(x) = 3x^{458} + 2x^{103} + 4$ in $\mathbb{Z}_5[x]$.
 - (a) Find the remainder when $f(x)$ is divided by $x - 2$.
 - (b) Find the remainder when $f(x)$ is divided by $x + 1$

2. Find $\alpha \in \mathbb{R}$ such that $f(\alpha) = 0$.
 - (a) $f(x) = 4x + 2 \in \mathbb{Z}_7[x]$.
 - (c) $f(x) = 3x + 8 \in \mathbb{Z}_{13}[x]$.

3. Fix $x^2 \in \mathbb{Z}_3[x]$ and define $\langle x^2 \rangle = \{x^2 \cdot q(x) \mid q(x) \in \mathbb{Z}_3[x]\}$.
 - (a) Verify whether or not / why $f(x) = x^6 + 2x$, $g(x) = 2x^7 + x^5 + 2x^4$ is contained / not contained in $\langle x^2 \rangle$.
 - (b) List three more elements of $\mathbb{Z}_3[x]$ that are contained in $\langle x^2 \rangle$.
 - (c) List three more elements of $\mathbb{Z}_3[x]$ that are not contained in $\langle x^2 \rangle$

4. In the quotient ring $\mathbb{Z}_3[x]/\langle x^2 \rangle$, explain why
 - (a) $(x + 1) + \langle x^2 \rangle \neq 2x + \langle x^2 \rangle$; (b) $(x + 2) + \langle x^2 \rangle \neq (2x + 2) + \langle x^2 \rangle$.

5. Prove that $\mathbb{Z}[\sqrt{3}i] = \{a + b\sqrt{3}i : a, b \in \mathbb{Z}\}$ is an integral domain. Verify whether the polynomial $f(x) = 5x^5 - 6x^4 - 3x^2 + 9x - 15$ is irreducible over rational numbers.